Example 2.3

A spinner of radius 7.5 *cm* is divided into 6 equal sectors. Find the area of each of the sectors.

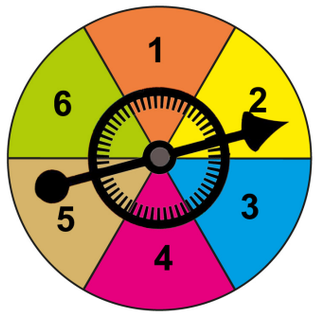


Fig. 2.15

Solution:

Radius, $r = 7.5\ cm$ and $n = 6$.

Area of each of the sectors, $A = \frac{1}{n} \times \pi r^2\ sq.\ units$

$$= \frac{1}{6} \times \pi \times 7.5 \times 7.5$$

$$= 9.375\pi\ sq.\ cm$$

Example 2.4

Kamalesh has a dining table, circular in shape of radius 70 *cm* whereas Tharun has a circular quadrant dining table of radius 140 *cm*. Whose dining table has a greater area? $\left(\pi = \frac{22}{7}\right)$


Fig. 2.16

Solution:

Area of the dining table with Kamalesh $= \pi r^2\ sq.\ units$

$$= \frac{22}{7} \times 70 \times 70$$

$A = 15400\ sq.cm$ (approximately.)


Fig. 2.17

Area of the circular quadrant dining table with Tharun

$$= \frac{1}{4}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 140 \times 140$$

$A = 15400\ sq.cm$ (approximately.)

We find that, the area of the dining tables of both of them have the same area.

Think



If the radius of a circle is doubled, what will happen to the area of the new circle so formed?

Example 2.5

Four identical medals, each of diameter 7*cm* are placed as shown in Fig. 2.18. Find the area of the shaded region between the medals. $\left(\pi = \frac{22}{7}\right)$


Fig. 2.18

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